Introduction to Cryptography

Lecture 8(B): Polyalphabetic Ciphers
Vigenère Cipher, Vernam Cipher and One-Time Pad

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Why Polyalphabetic Ciphers?

Problem

Simple monoalphabetic substitution preserves the statistical properties of the plaintext language.

- One plaintext letter maps to one ciphertext letter.
- Letter frequency analysis can break the cipher.
- We need different substitution rules at different positions.

Solution

Use multiple substitution alphabets while encrypting the message.

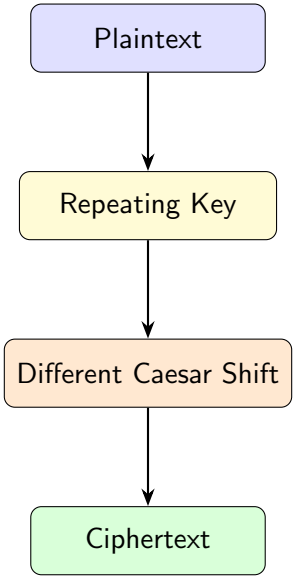
Polyalphabetic Substitution Cipher

Definition

A polyalphabetic substitution cipher uses a set of related monoalphabetic substitution rules, with the key deciding which rule is used at each position.

- Different letters of the key produce different shifts.
- Same plaintext letter may produce different ciphertext letters.
- Letter frequency information is partially hidden.
- Vigenère cipher is a famous example.

Polyalphabetic Cipher Concept



Vigenère Cipher

Definition

The Vigenère cipher is a polyalphabetic substitution cipher that uses the 26 Caesar ciphers with shifts from 0 to 25.

- A keyword is repeated until it matches the length of plaintext.
- Each key letter defines a Caesar shift.
- Encryption is done letter by letter.
- It is stronger than monoalphabetic substitution.

Letter-to-Number Mapping

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Vigenère Encryption Formula

Let plaintext be:

$$P = p_0, p_1, \ldots, p_{n-1}$$

Let key be:

$$K = k_0, k_1, \ldots, k_{m-1}$$

Encryption:

$$C_i = (p_i + k_{i \bmod m}) \bmod 26$$

Meaning

The key repeats after every m characters.

Vigenère Decryption Formula

Decryption:

$$p_i = (C_i - k_{i \bmod m}) \bmod 26$$

where:

- p_i is plaintext value.
- C_i is ciphertext value.
- $k_{i \bmod m}$ is the repeating key value.
- All operations are modulo 26.

Vigenère Example

Plaintext:

```
wearediscoveredsaveyourself
```

Keyword:

```
deceptive
```

Repeated key:

```
deceptivedeceptivedeceptive
```

Ciphertext:

```
ZICVTWQNGRZGVTWAVZHCQYGLMGJ
```

Solved Vigenère Numerical 1

Encrypt plaintext:

`weare`

using key:

`deceptive`

Use only first five key letters:

`decep`

Plaintext	*w*	*e*	*a*	*r*	*e*
P value	22	4	0	17	4
Key	*d*	*e*	*c*	*e*	*p*
K value	3	4	2	4	15

Solved Vigenère Numerical 1: Solution

$$C_i = (p_i + k_i) \bmod 26$$

P value	22	4	0	17	4
K value	3	4	2	4	15
Sum	25	8	2	21	19
Cipher	Z	I	C	V	T

Ciphertext

`weare` $\rightarrow$ `ZICVT`

Solved Vigenère Numerical 2

Encrypt plaintext:

`attack`

using key:

`lemon`

Repeat key:

`lemonl`

Plaintext	*a*	*t*	*t*	*a*	*c*	*k*
P value	0	19	19	0	2	10
Key	*l*	*e*	*m*	*o*	*n*	*l*
K value	11	4	12	14	13	11

Solved Vigenère Numerical 2: Solution

P value	0	19	19	0	2	10
K value	11	4	12	14	13	11
Sum mod 26	11	23	5	14	15	21
Cipher	*L*	*X*	*F*	*O*	*P*	*V*

Ciphertext

`attack` $\rightarrow$ `LXFOPV`

Solved Vigenère Decryption

Decrypt:

`LXFOPV`

using key:

`lemonl`

Use:

$$p_i = (C_i - k_i) \bmod 26$$

Cipher	L	X	F	O	P	V
C value	11	23	5	14	15	21
K value	11	4	12	14	13	11

Solved Vigenère Decryption: Solution

C value	11	23	5	14	15	21
K value	11	4	12	14	13	11
Difference mod 26	0	19	19	0	2	10
Plain	*a*	*t*	*t*	*a*	*c*	*k*

Plaintext

`LXFOPV` $\rightarrow$ `attack`

Strength of Vigenère Cipher

- Same plaintext letter can produce different ciphertext letters.
- Frequency information is more hidden than in monoalphabetic ciphers.
- It uses multiple Caesar ciphers.
- Longer keywords generally increase resistance.

Limitation

If the keyword repeats, patterns may still appear in the ciphertext.

Cryptanalysis of Vigenère Cipher

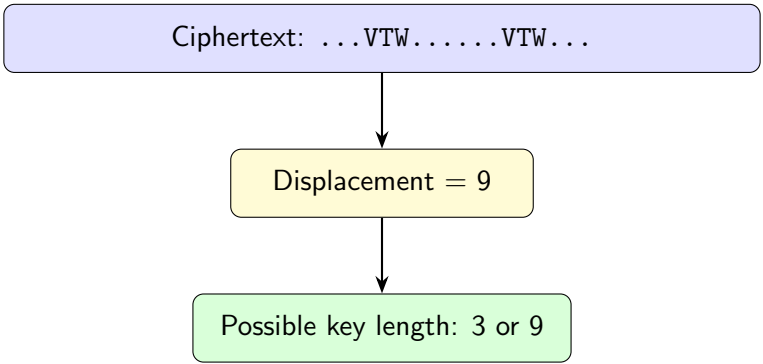
Key idea:

- If identical plaintext sequences occur at distances that are multiples of the keyword length, they may produce identical ciphertext sequences.
- Repeated ciphertext sequences can reveal possible keyword length.
- Once keyword length is known, the cipher can be treated as multiple Caesar ciphers.

Example

Repeated sequence `VTW` at displacement 9 may suggest key length 3 or 9.

Repeated Sequence Analysis



Autokey System

Idea

Instead of repeating the keyword, concatenate the keyword with the plaintext itself to create a running key.

Example:

Key: `deceptivewearediscoveredsav`

Plaintext: `wearediscoveredsaveyourself`

Ciphertext: `ZICVTWQNGKZEIIGASXSTSLVVWLA`

Limitation of Autokey

- It avoids simple periodic repetition of the keyword.
- However, the key contains plaintext characters.
- The key and plaintext share statistical properties.
- Statistical cryptanalysis may still be possible.

Observation

Autokey improves Vigenère but does not provide perfect secrecy.

Vernam Cipher

Definition

The Vernam cipher operates on binary data and combines plaintext bits with key bits using XOR.

$$c_i = p_i \oplus k_i$$

where:

- p_i = plaintext bit
- k_i = key bit
- c_i = ciphertext bit
- $\oplus$ = XOR operation

XOR Operation

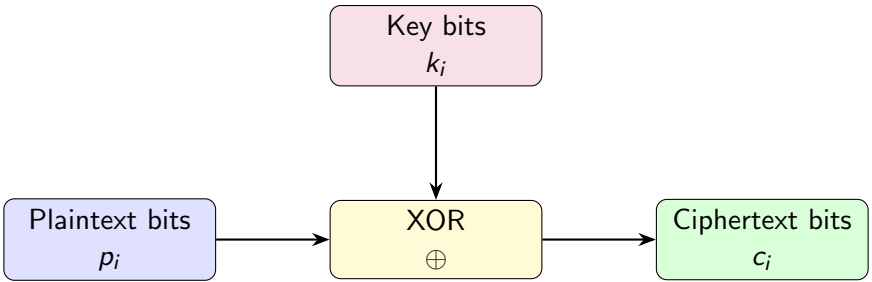
p_i	k_i	$c_i = p_i \oplus k_i$
0	0	0
0	1	1
1	0	1
1	1	0

Property

XOR is its own inverse.

$$p_i = c_i \oplus k_i$$

Vernam Cipher Model



Solved Vernam Numerical

Encrypt:

$$P = 101101$$

using key:

$$K = 011010$$

$$C = P \oplus K$$

P	1	0	1	1	0	1
K	0	1	1	0	1	0
C	1	1	0	1	1	1

Ciphertext

$$C = 110111$$

Solved Vernam Decryption

Decrypt:

$$C = 110111$$

using key:

$$K = 011010$$

$$P = C \oplus K$$

C	1	1	0	1	1	1
K	0	1	1	0	1	0
P	1	0	1	1	0	1

Plaintext

$$P = 101101$$

One-Time Pad

Definition

A one-time pad uses a truly random key that is as long as the plaintext, used only once, and then discarded.

- Key length equals message length.
- Key must be truly random.
- Key must never be reused.
- Key must remain secret.
- It provides perfect secrecy in theory.

One-Time Pad Conditions

For perfect secrecy, the one-time pad requires:

1. The key must be truly random.
2. The key must be at least as long as the plaintext.
3. The key must be used only once.
4. The key must be known only to sender and receiver.

Warning

Reusing a one-time pad key destroys its security.

Why One-Time Pad is Unbreakable

- Ciphertext has no statistical relationship with plaintext.
- For any possible plaintext, there exists a key that produces it.
- Exhaustive key search gives many meaningful plaintexts.
- The cryptanalyst cannot determine which plaintext is correct.

Key Point

Its security comes entirely from the randomness and one-time use of the key.

One-Time Pad Example Concept

The same ciphertext may decrypt to different meaningful plaintexts if different keys are used.

$$\texttt{ciphertext} + \texttt{key 1} \rightarrow \texttt{message 1}$$

$$\texttt{ciphertext} + \texttt{key 2} \rightarrow \texttt{message 2}$$

Conclusion

Without knowing the correct random key, the attacker cannot decide which plaintext is correct.

Practical Problems of One-Time Pad

1. **Random key generation:** Large quantities of truly random keys are difficult to generate.
2. **Key distribution:** A key as long as the message must be securely shared.
3. **Key storage:** Large keys must be stored securely.
4. **Key reuse risk:** Reusing a key makes the system vulnerable.

Practical Use

Useful mainly for low-bandwidth communication requiring very high security.

Comparison of Ciphers

Cipher	Key Type	Strength	Weakness
Vigenère	Repeating keyword	Hides simple frequency	Repeated key patterns
Autokey	Keyword + plaintext	Reduces periodicity	Statistical leakage
Vernam	Binary key stream	Simple XOR operation	Repeating key vulnerable
One-Time Pad	Truly random one-time key	Perfect secrecy	Key distribution

Class Practice 1

Encrypt:

`meet`

using Vigenère key:

`key`

Repeat key:

`keyk`

Task

Use:

$$C_i = (p_i + k_i) \bmod 26$$

Class Practice 1: Answer

Plain	m	e	e	t
P value	12	4	4	19
Key	k	e	y	k
K value	10	4	24	10
C value	22	8	2	3
Cipher	W	I	C	D

Ciphertext

$$\texttt{meet} \rightarrow \texttt{WICD}$$

Class Practice 2

Encrypt using Vernam cipher:

$$P = 1001101$$

$$K = 0101011$$

Task

Find:

$$C = P \oplus K$$

Class Practice 2: Answer

P	1	0	0	1	1	0	1
K	0	1	0	1	0	1	1
C	1	1	0	0	1	1	0

Ciphertext

$$C = 1100110$$

Quick Quiz

1. Define polyalphabetic substitution cipher.
2. Why is Vigenère stronger than monoalphabetic substitution?
3. Write the Vigenère encryption formula.
4. Encrypt `attack` using key `lemon`.
5. What is the role of repeated sequences in Vigenère cryptanalysis?
6. Write the Vernam cipher formula.
7. Why is XOR used in Vernam cipher?
8. State the conditions required for a one-time pad to be perfectly secure.

References

- William Stallings, *Cryptography and Network Security: Principles and Practice*.
- Douglas R. Stinson, *Cryptography: Theory and Practice*.
- Menezes, van Oorschot, and Vanstone, *Handbook of Applied Cryptography*.

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- Clarity of concepts
- Quality of examples
- Pace of the lecture
- Suggestions for improvement

Thank you for your valuable response.



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